

Technical & Ethical Report

AI-Enabled Low-Cost Real-Time Mine Subsidence Monitoring, Prediction, and Early Warning System

Sponsoring Ministry: Ministry of Coal (Coal India Limited) | Problem Statement ID: SIH26025

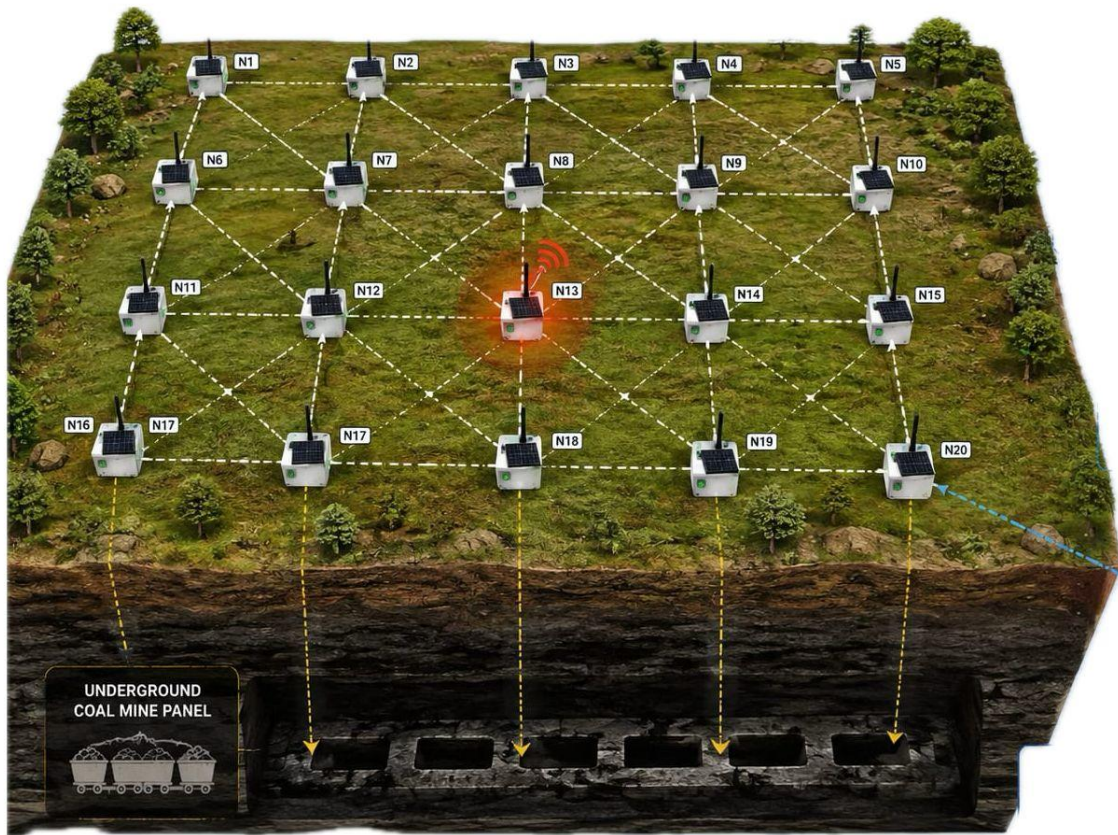
Target Audience: Sponsoring Ministry, Coal India Limited (CIL), Geotechnical Engineers, System Architects, and Academic Evaluators

GREEN EYES

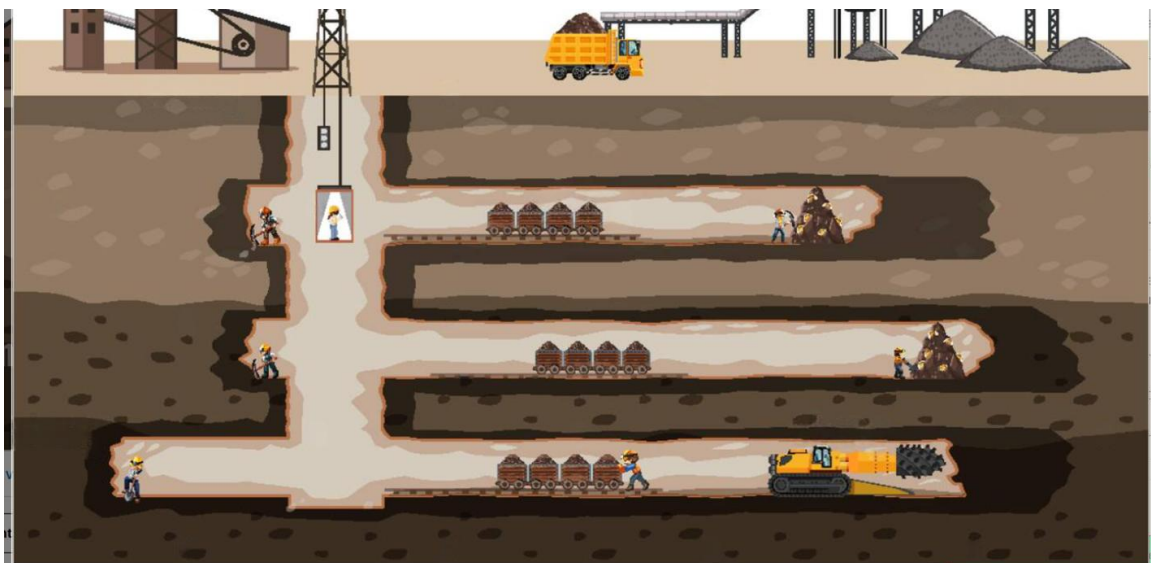
Executive Summary

Surface land subsidence represents the final environmental, structural, and public-safety consequence of underground mining excavations. When coal pillars are systematically extracted during the depillaring phase of bord-and-pillar mining, or when large blocks of coal are removed in mechanized longwall mining, massive subsurface voids (the goaf) are created. Under the relentless force of gravity and geomechanical stress, the overlying rock layers fracture, buckle, and collapse, propagating a progressive failure upward through the stratigraphic column until it manifests as a visible depression or fracture zone on the ground surface.

This comprehensive master report details the architecture of a surface-based, AI-enabled, low-cost, real-time mine subsidence monitoring, prediction, and early warning system. By placing distributed sensor nodes on the ground surface, the physical constraints, signal blockage, and extreme safety restrictions (like flameproof certifications) of subterranean spaces are bypassed, establishing a direct line-of-sight to the terrain undergoing deformation.



Surface-to-Subterranean Interaction



Underground Mining Cross-Section

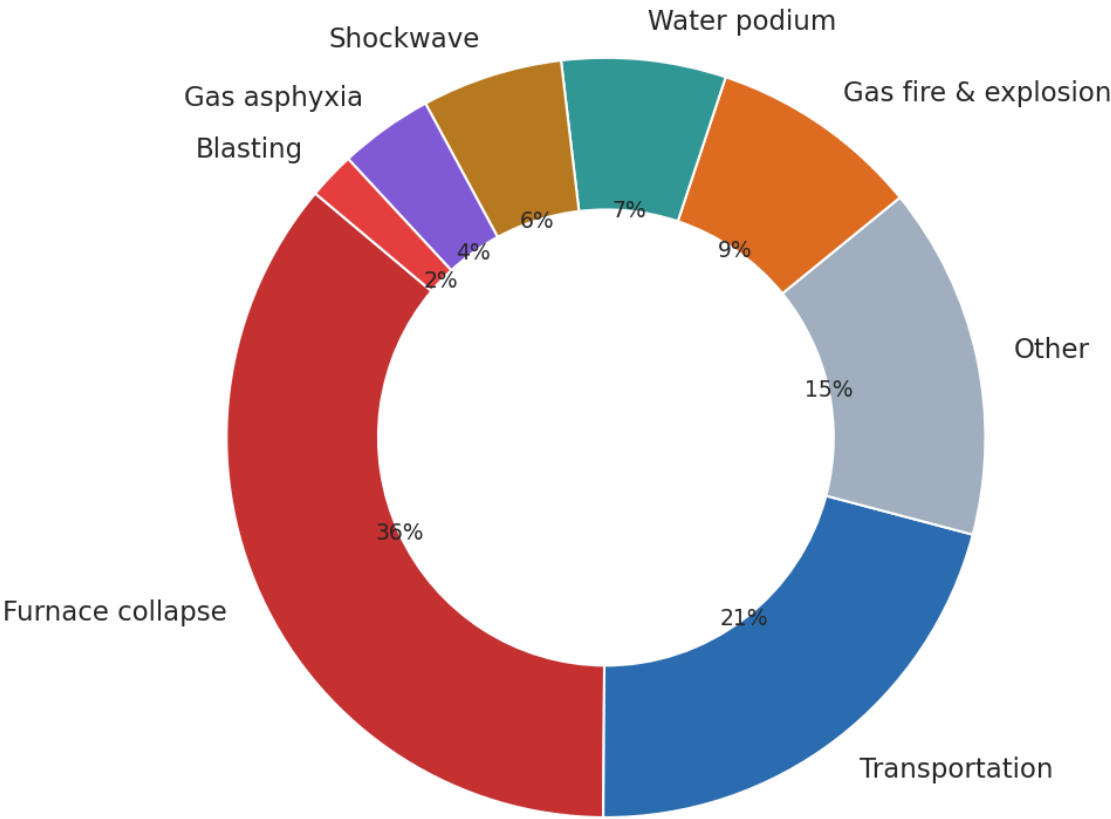
1. Project Background and Socioeconomic Necessity

Underground coal mining remains a cornerstone of India's energy security, but its surface consequences present severe environmental, legal, and safety hazards. The system's design must prove high feasibility and commercial viability for Coal India Limited and its subsidiaries to successfully defend this proposal before the Smart India Hackathon jury.

1.1 Statistical Profile of Mining Hazards

Roof collapses and ground failures are historically the most lethal categories of accidents in Indian mining operations. Based on historical mine accident data compiled by national safety agencies, furnace and roof collapses represent 36% of all mining accidents, vastly outstripping hazards like gas explosions (9%), transportation failures (21%), and water inundation (7%).

Statistical Distribution of Underground Coal Mining Hazards in India



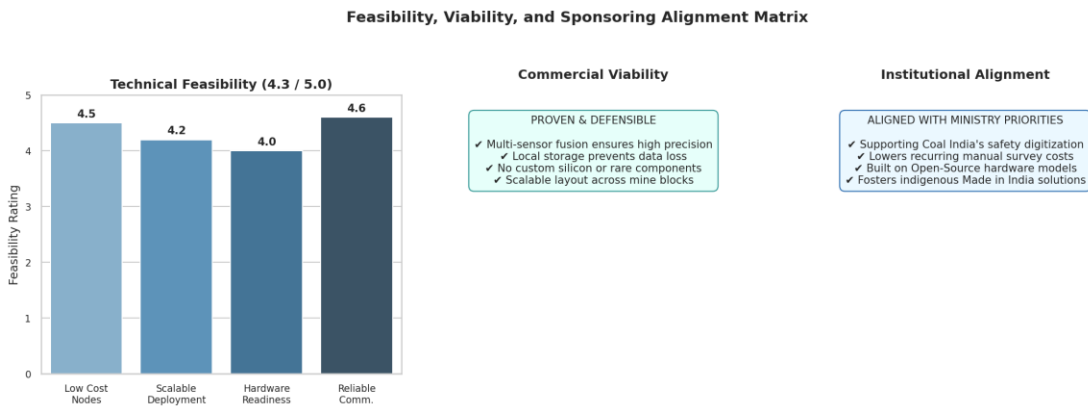
Historical Breakdown of Indian Coal Mining Safety Threats

1.2 Institutional Alignment and Project Feasibility

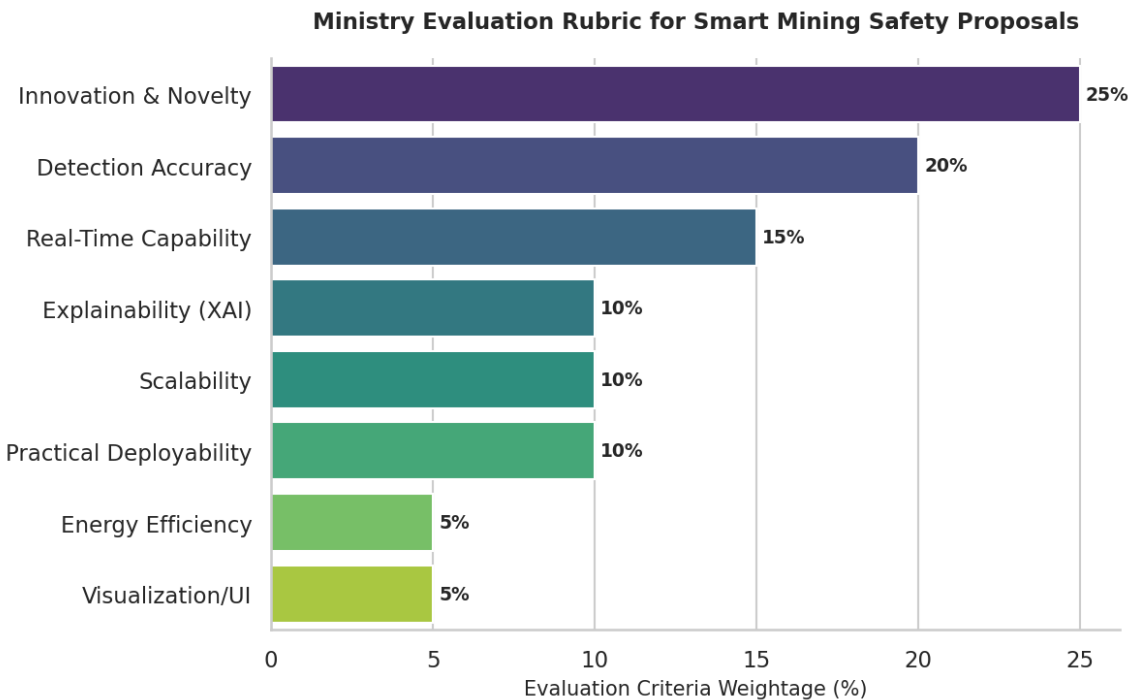
Feasibility (4.3 out of 5): Achieved by utilizing off-the-shelf, modular hardware components (such as the ESP32 and MPU-6050) that require zero custom silicon fabrication.

Viability: Supported by a localized data buffering system using non-volatile flash and micro-SD storage to maintain zero-loss telemetry during cellular outages.

Business Potential: Delivers massive operational cost savings by replacing slow and hazardous manual surveying with a continuous 24/7 monitoring grid.



Project Feasibility, Viability, and Sponsoring Alignment Matrix



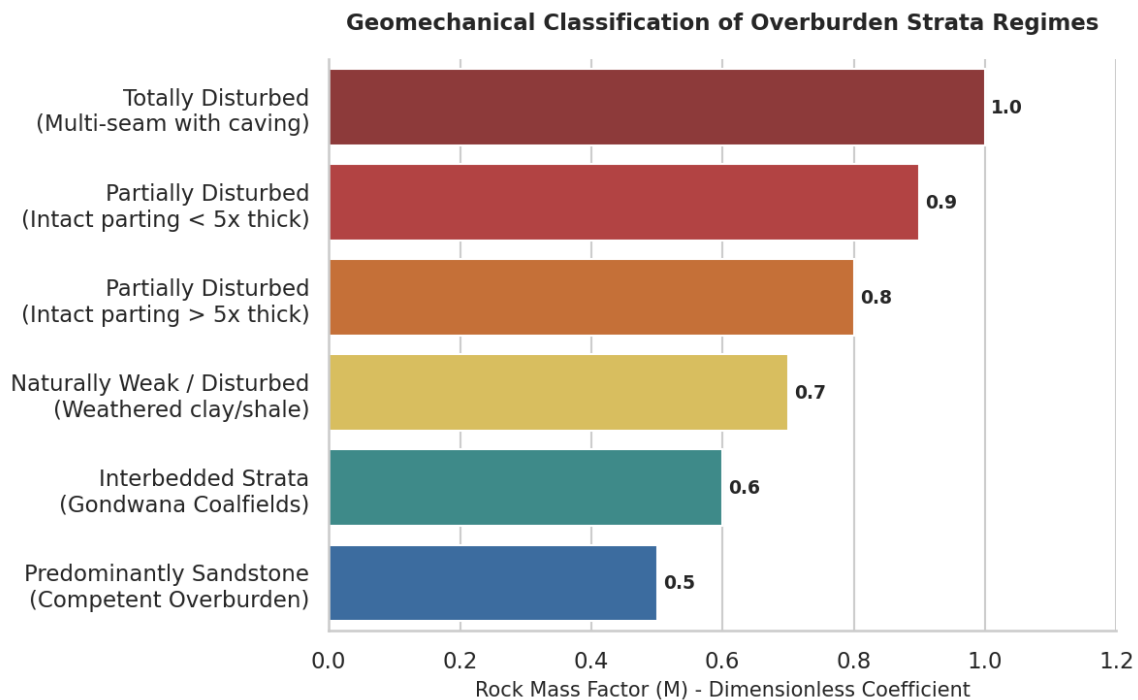
2. Geomechanical and Strata Engineering Foundations

An effective AI predictive system must be physically constrained by empirical formulations of rock mechanics and subsidence engineering, deeply rooted in research compiled by bodies like the CSIR-CIMFR and Naresh C. Saxena's Subsidence Management Handbook.

2.1 The Rock Mass Factor

The macro-geological behavior of the overburden dictates how subsurface failures translate into surface troughs. We categorize the overburden using the Rock Mass Factor (M):

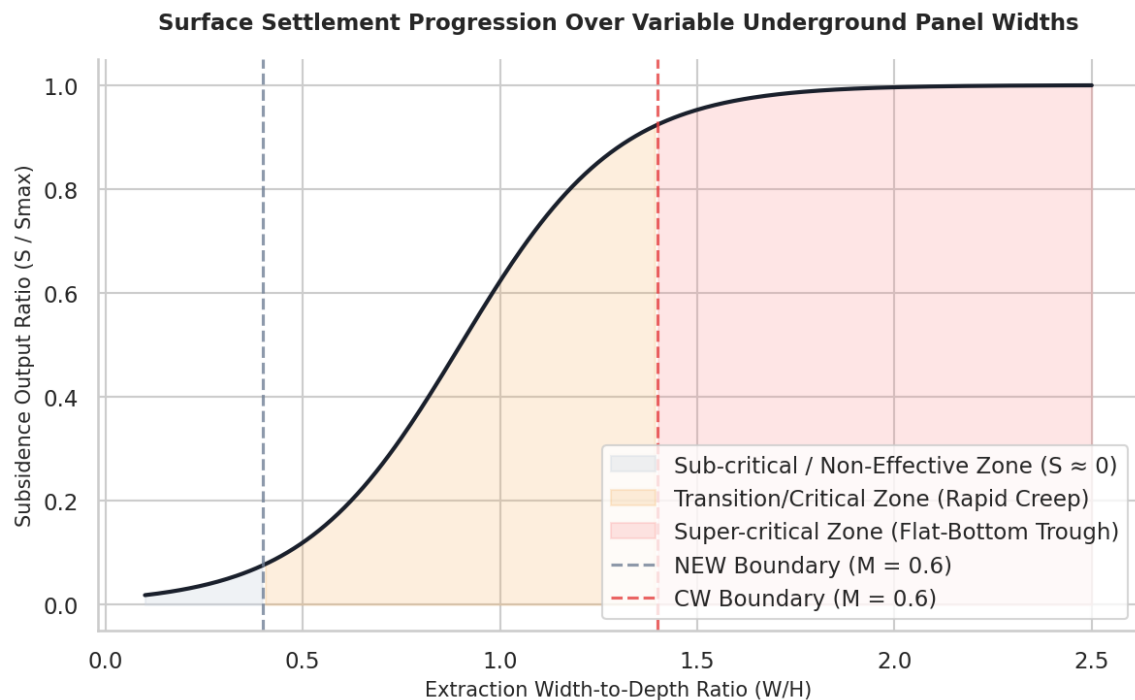
- M = 1.0 (Totally Disturbed Strata): Found in multi-seam slicing with active caving.
- M = 0.6 (Interbedded Sandstones and Shales): The standard stratigraphic column of the Gondwana coalfields.
- M = 0.5 (Predominantly Sandstone Column): Highly competent beds that exhibit bridging behavior and sudden fracturing upon failure.



Rock Mass Factor Strata Classification Chart

2.2 Critical Width-Depth Extractions

- Non-Effective Width-Depth Ratio (NEW): The critical extraction span below which zero vertical or horizontal displacement can be detected on the surface, calculated as 1 minus the Rock Mass Factor.
- Critical Width-Depth Ratio (CW): The threshold at which the vertical subsidence at the center of the surface trough reaches its absolute maximum possible value, modeled as 2 minus the Rock Mass Factor.

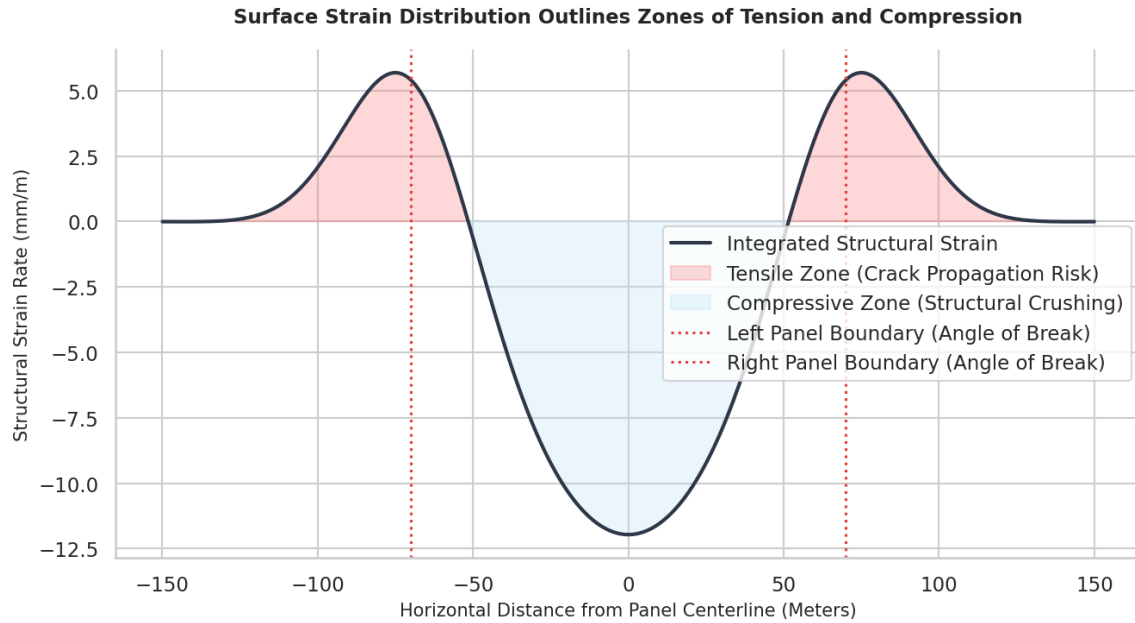


Surface Settlement Zones and Extraction Width-Depth Ratios

2.3 Horizontal Ground Strain and Shear Distributions

Vertical settlement is accompanied by horizontal displacement, creating dangerous shear and tensile stresses.

- Compressive Zone: Over the center of the panel, causing soil buckling.
- Tensile Zone: Over the outer boundaries aligning with the Angle of Break, pulling soil apart and creating high tension that initiates open surface cracks.



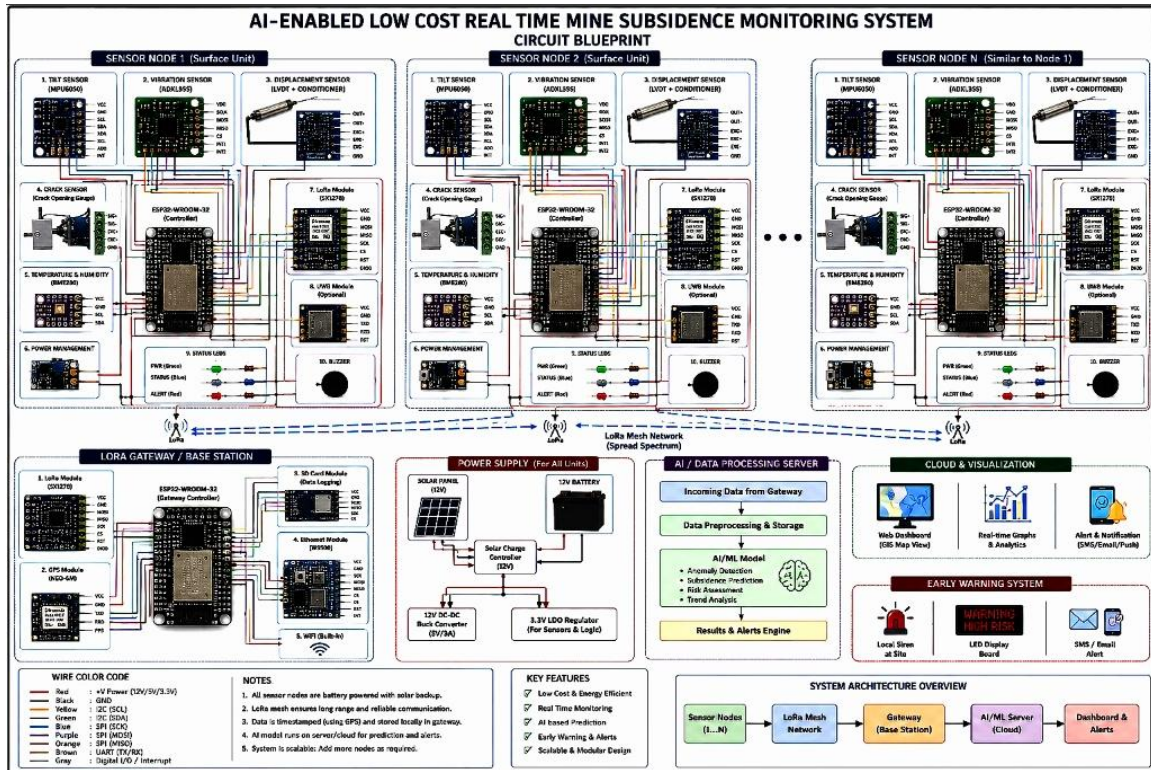
Horizontal Strain, Shear, and Tension Profile across the Subsidence Basin

3. Localized Wireless Surface Mesh Network (Hardware Architecture)

To capture geomechanical shifts in real time without prohibitive costs, we deploy a localized, self-healing wireless surface mesh network.

3.1 Edge Node Hardware and Pinout Blueprint

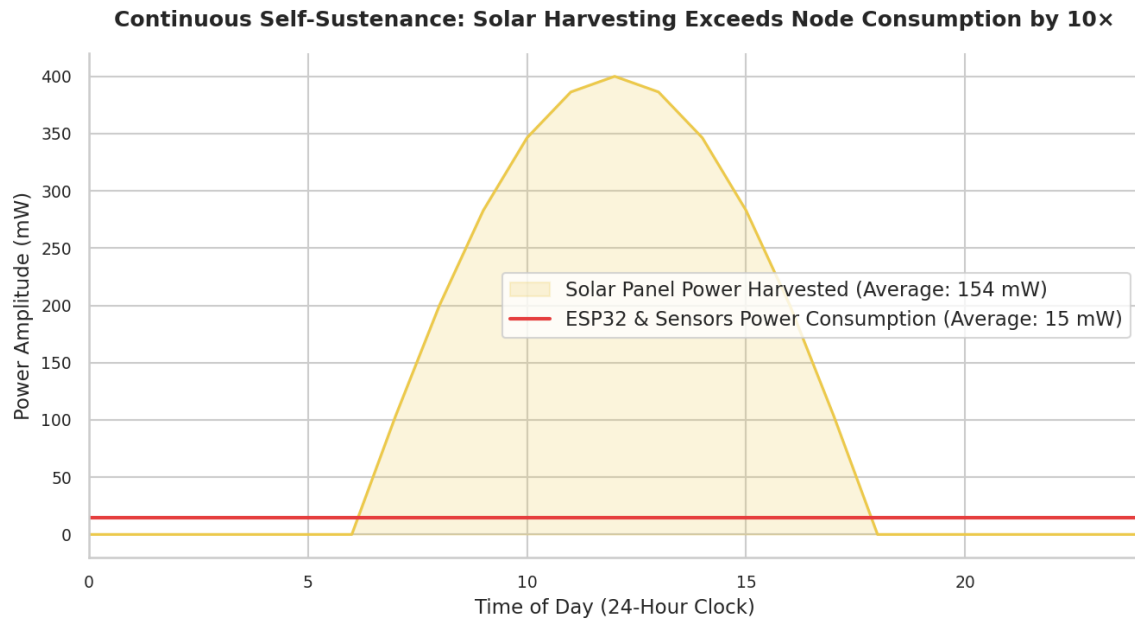
Each node is built using the dual-core, 32-bit ESP32 microcontroller integrated with high-precision, low-cost MEMS sensors. The hardware layout integrates the ESP32 controller, MPU-6050 (tilt), ADXL355 (vibration), LVDT (linear displacement), BME280 (temperature/humidity), and SX1278 LoRa transceiver module.



Detailed Node Circuit Design

3.2 24-Hour Solar Energy Harvesting Cycle

Nodes utilize a 5W polycrystalline solar panel paired with dual Lithium-Iron Phosphate (LiFePO₄) cells to ensure permanent operation without external grid connections. The deep-sleep duty cycling keeps power consumption low, allowing the battery to remain self-sustaining even through several days of monsoon cloud cover.

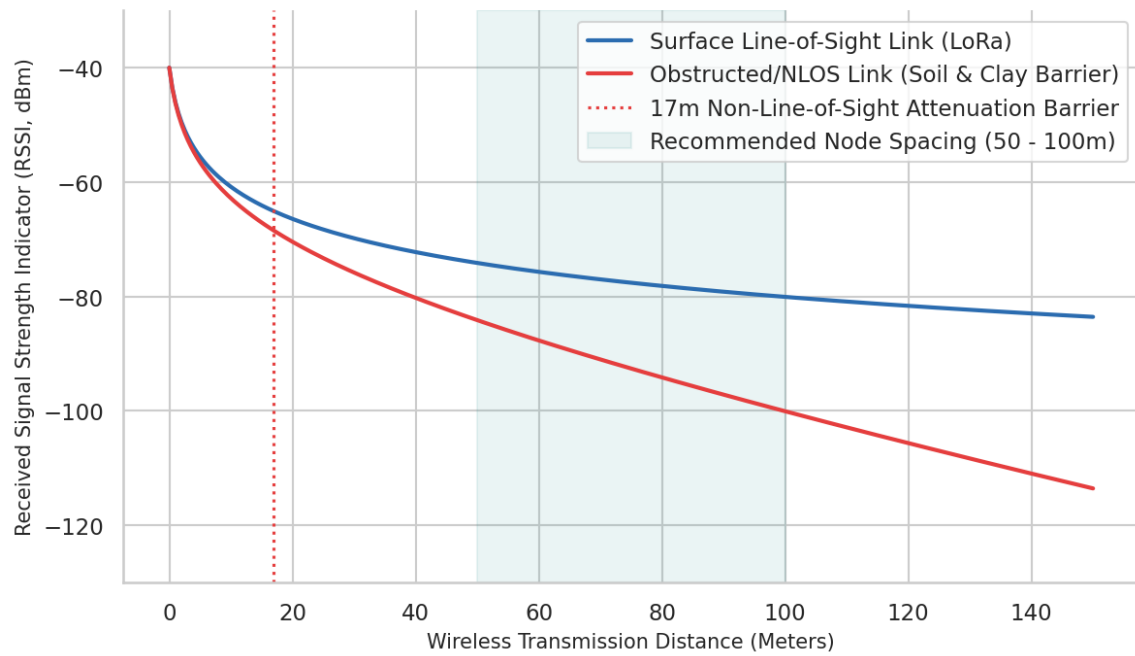


24-Hour Solar Power Harvesting and Battery State-of-Charge Cycle

3.3 Wireless Spacing, Attenuation, and Link Budget

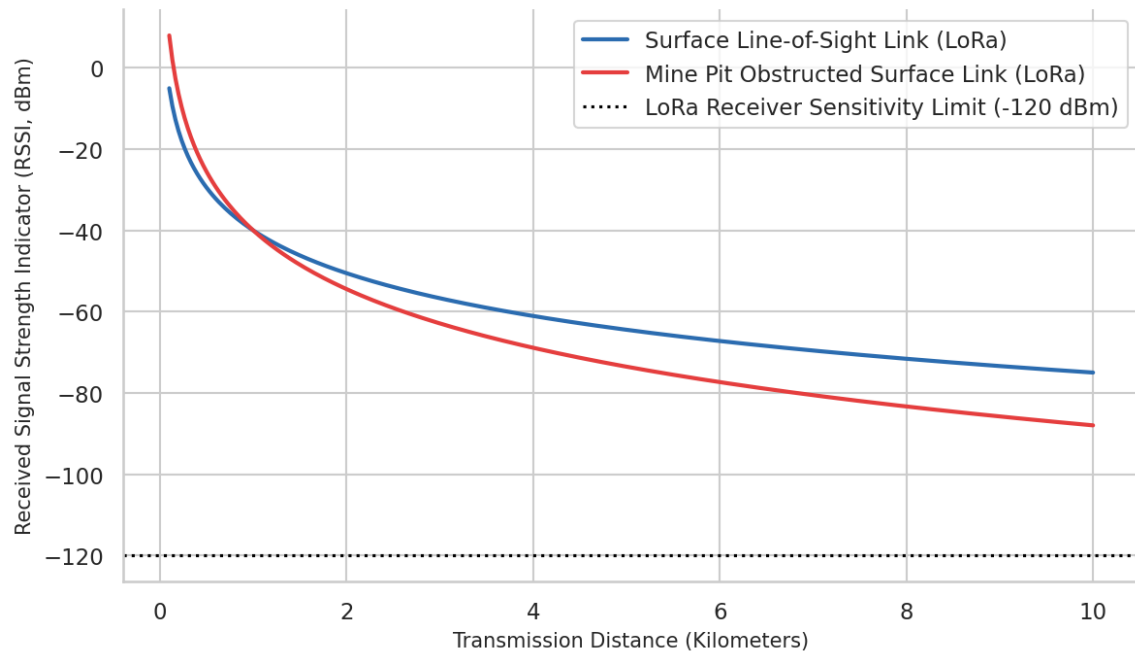
While underground NLOS (non-line-of-sight) signals drop completely within 17 meters due to electromagnetic absorption by wet coal, surface mesh spacing is optimized at 50 to 100 meters. LoRa Chirp Spread Spectrum (CSS) modulation successfully decodes telemetry packets even when signal strength drops near the industrial noise floor of -120 dBm.

Surface Deployment Bypasses Subterranean Signal Attenuation Barriers



Wireless Signal Attenuation and Spacing Guidelines

LoRa Chirp Spread Spectrum Extends Range across Obstructed Mine Surfaces

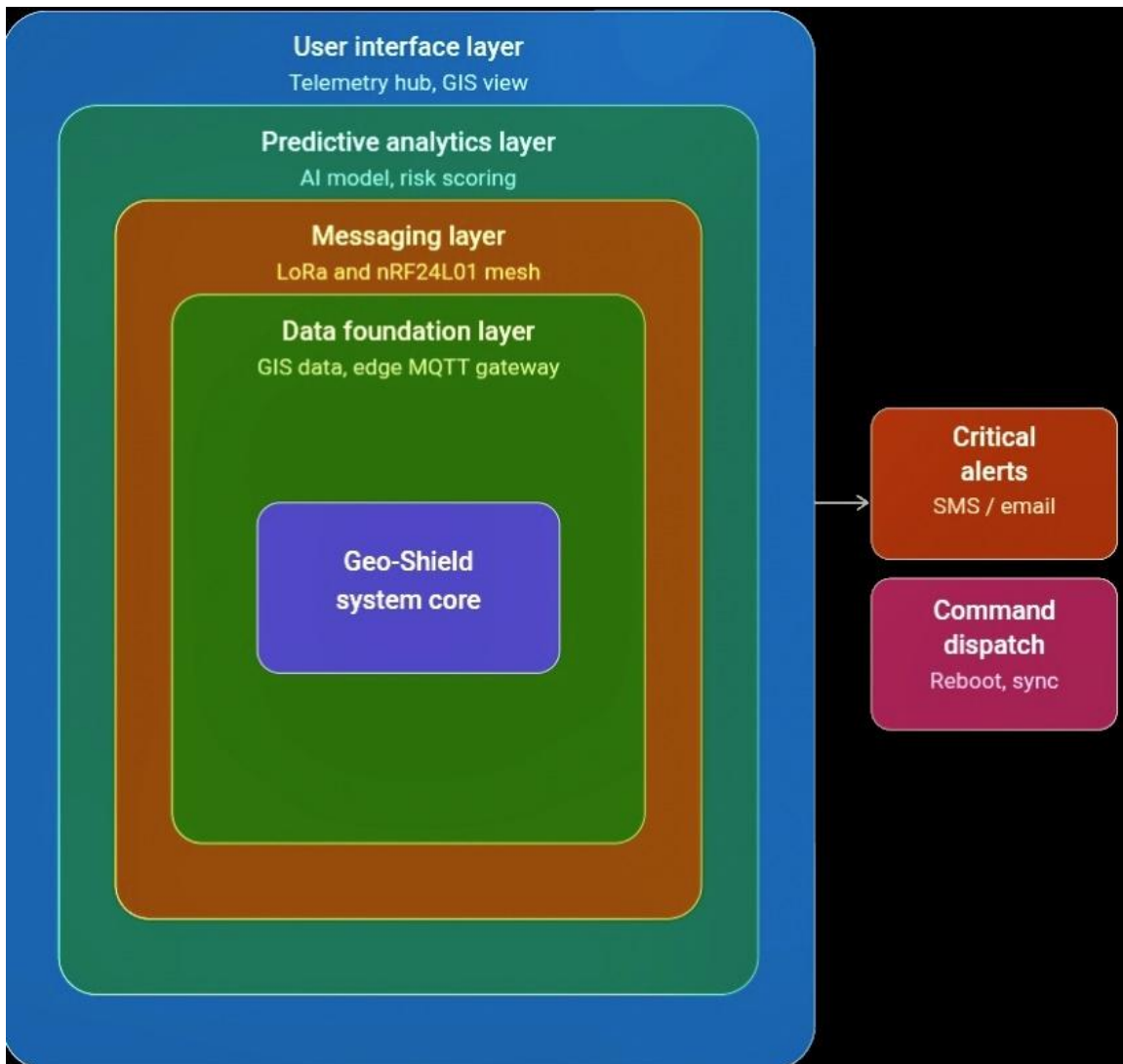


LoRa Signal RSSI and Path Loss over Distance

4. Edge-to-Cloud Telemetry & Communication Architecture

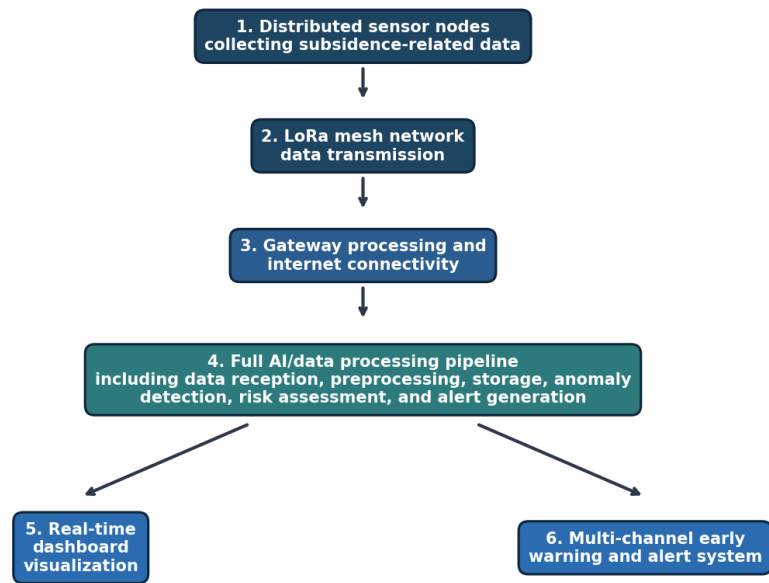
4.1 Automated Telemetry Pipeline

1. Distributed Sensing: ESP32 nodes sample tilt, vibration, and crack expansion.
2. LoRa Mesh Transmission: Packets route through adjacent nodes using self-healing routing.
3. Gateway Aggregation: The ESP32 base station receives and buffers data locally.
4. Cloud Synchronization: Pushes telemetry via cellular 4G or Wi-Fi.
5. AI Processing: Unsupervised anomaly detection and LSTM models evaluate risk.
6. Command and Warning: Triggers dashboards, sirens, and SMS alerts.



System Layer Architecture

SUBSIDENCE MONITORING DATA PIPELINE FLOWCHART

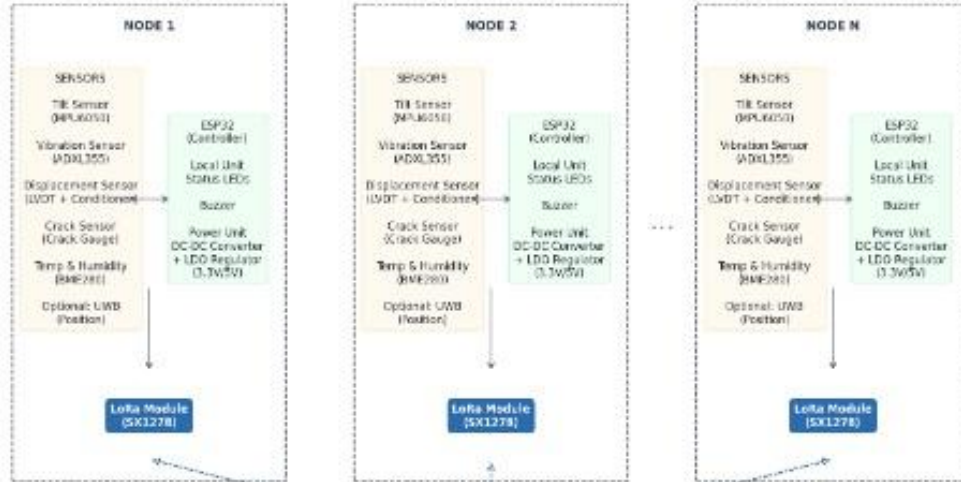


System Data Ingestion and Multi-Channel Alert Pipeline

4.2 Comprehensive System Flow

AI-ENABLED LOW COST REAL TIME MINE SUBSIDENCE MONITORING & EARLY WARNING SYSTEM - COMPLETE FLOW CHART

SENSOR NODES (DISTRIBUTED OVER MINE SURFACE)



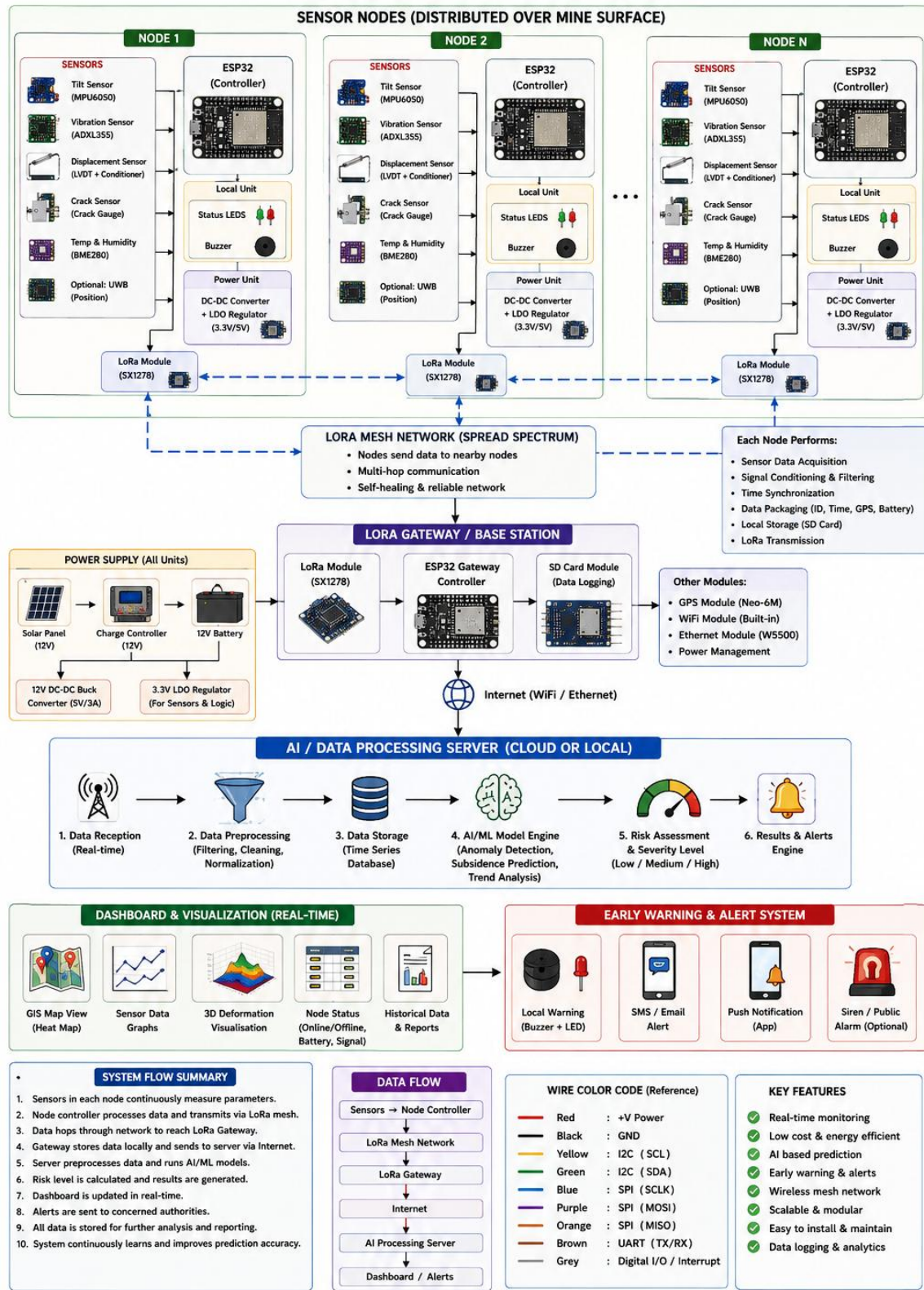
LoRa MESH NETWORK (SPREAD SPECTRUM)

- Nodes send data to nearby nodes
- Multi-hop communication
- Self-healing & reliable network



Complete Edge-to-Cloud Functional Flow

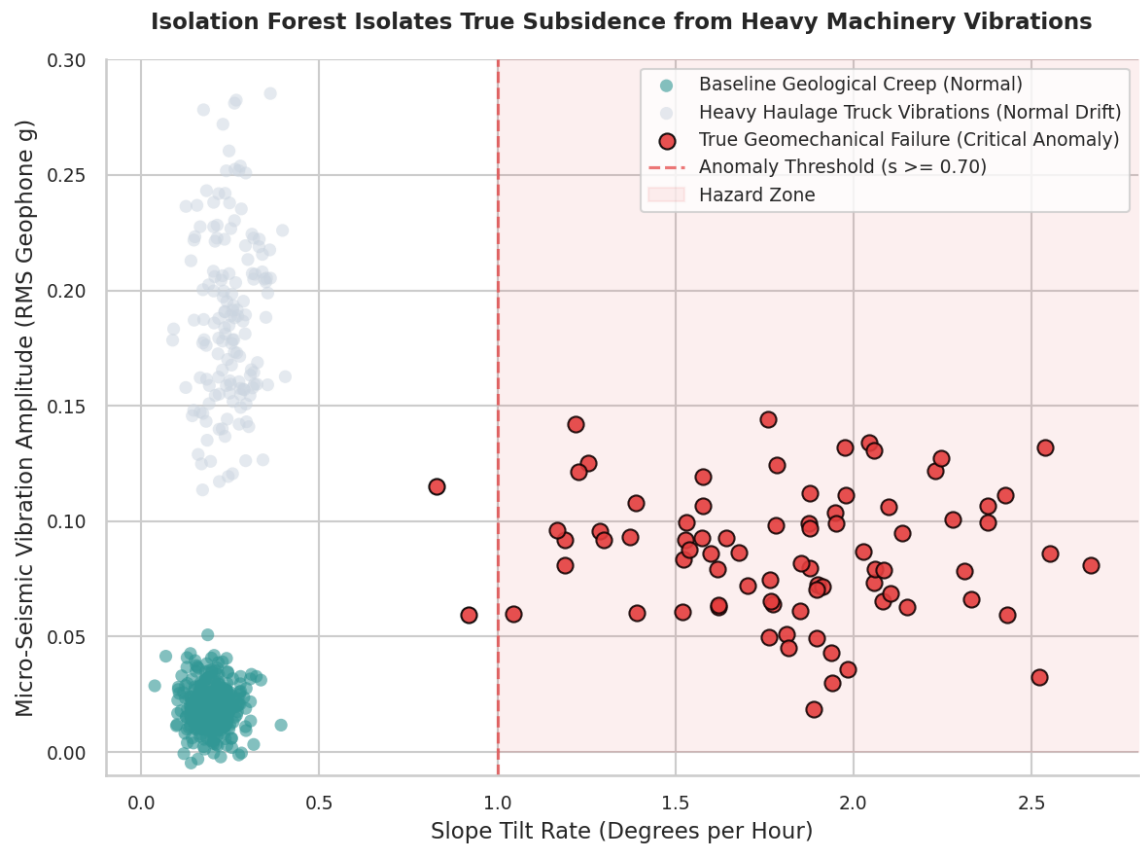
AI-ENABLED LOW COST REAL TIME MINE SUBSIDENCE MONITORING & EARLY WARNING SYSTEM – COMPLETE FLOW CHART



5. Dual-Track Artificial Intelligence Pipeline

Track A: Real-Time Anomaly Detection & Calibration

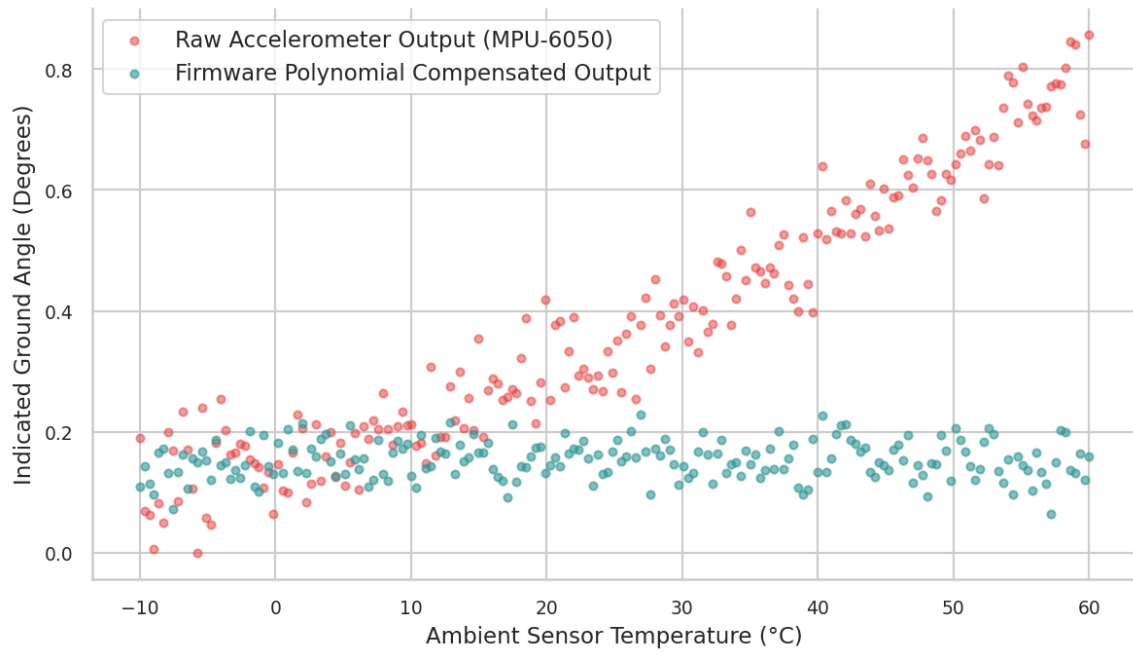
Unsupervised Anomaly Isolation using Isolation Forest: To prevent false alarms from heavy dump trucks or blasting, the Isolation Forest model fuses multi-axial tilt, angular velocity, and vibration amplitude to isolate genuine geomechanical ground shifts from repetitive machinery noise.



Isolation Forest Anomaly Classification and Decision Boundary

Dynamic Thermal Drift Calibration: Low-cost MEMS accelerometers are highly sensitive to seasonal temperature changes. The firmware utilizes co-located BME280 temperature readings to continuously correct the raw accelerometer output using a third-order calibration polynomial, flatlining thermal drift.

Firmware-Level Polynomial Calibration Corrects Accelerometer Temperature Drift

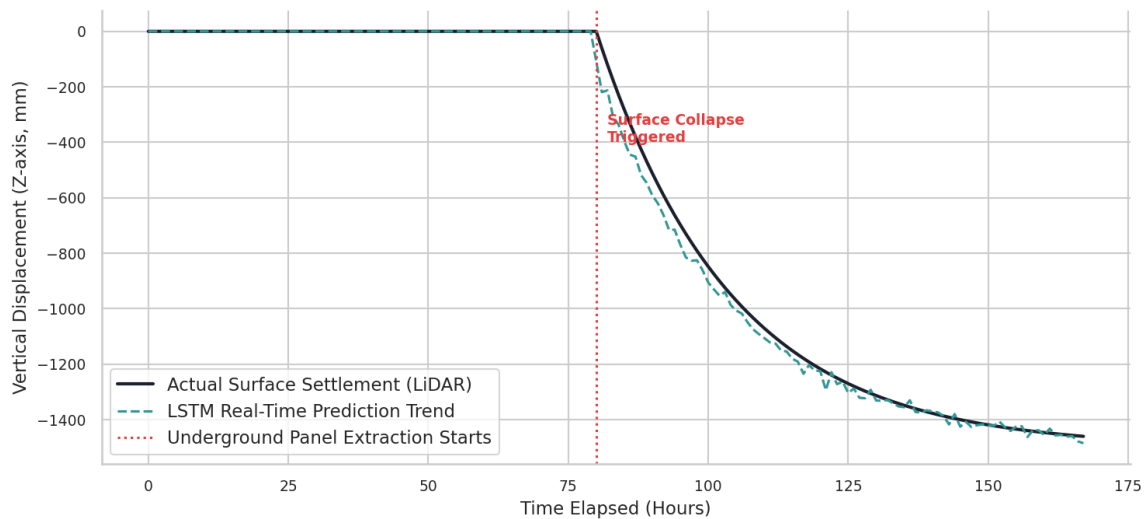


MPU-6050 Temperature Calibration and Compensation Curve

Track B: Long-Term Predictive Modeling

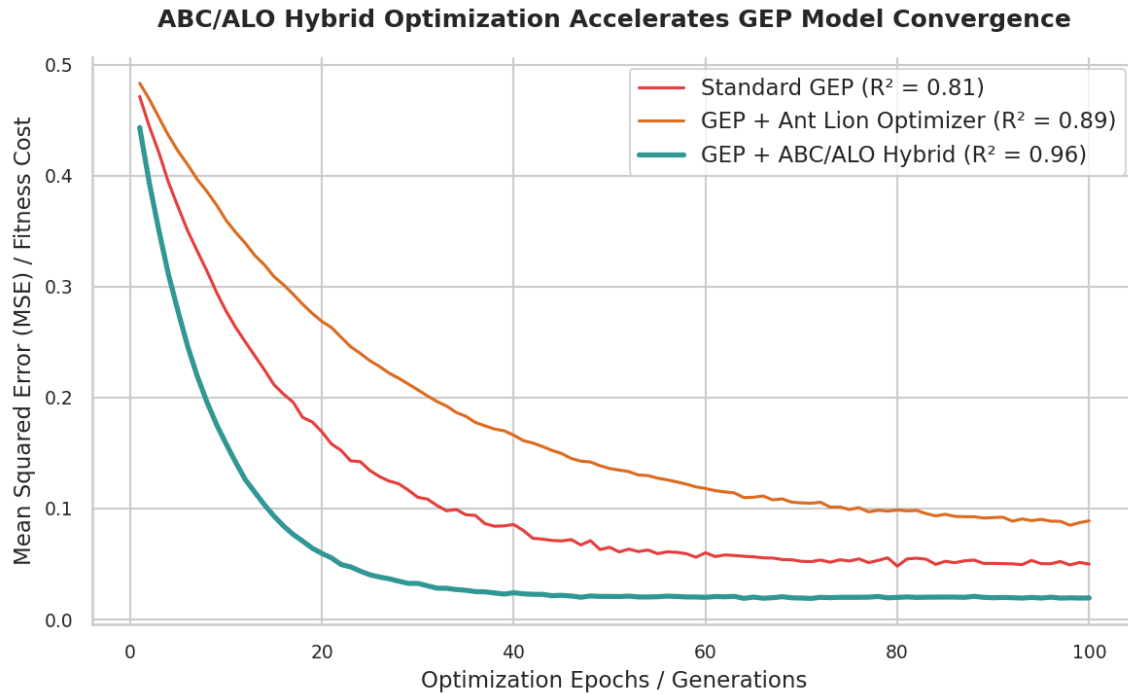
Time-Series Ground Creep Forecasting using LSTMs: The deep Long Short-Term Memory (LSTM) network analyzes historical sliding windows of surface displacement to predict vertical settlement and horizontal strain profiles 12 to 72 hours in advance of a failure.

Deep LSTM Predicts Post-Extraction Settlement Profile 12 Hours in Advance



LSTM Time-Series Forecasting vs. Actual Ground Settlement

Geomechanical Optimization using Evolutionary Programming: Gene Expression Programming (GEP), optimized through Artificial Bee Colony (ABC) and Ant Lion Optimizer (ALO) algorithms, achieves a correlation coefficient of 0.96 in predicting the final subsidence bowl under Gondwana stratigraphic columns.

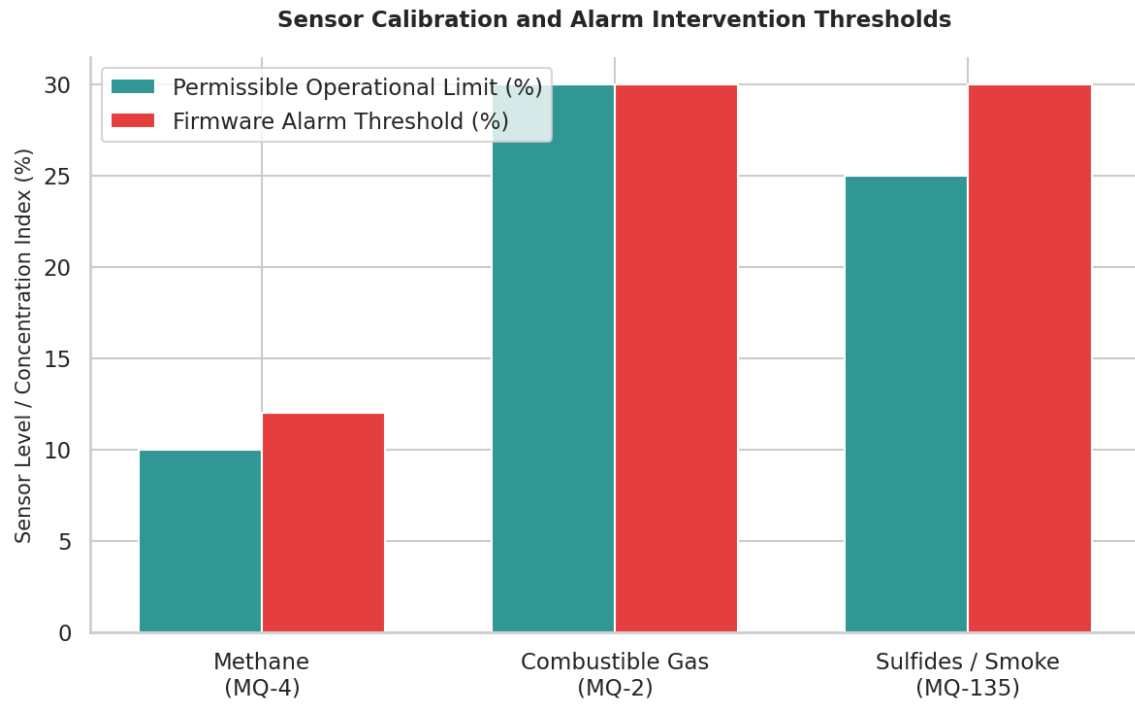


GEP Optimization Convergence Rate Comparison

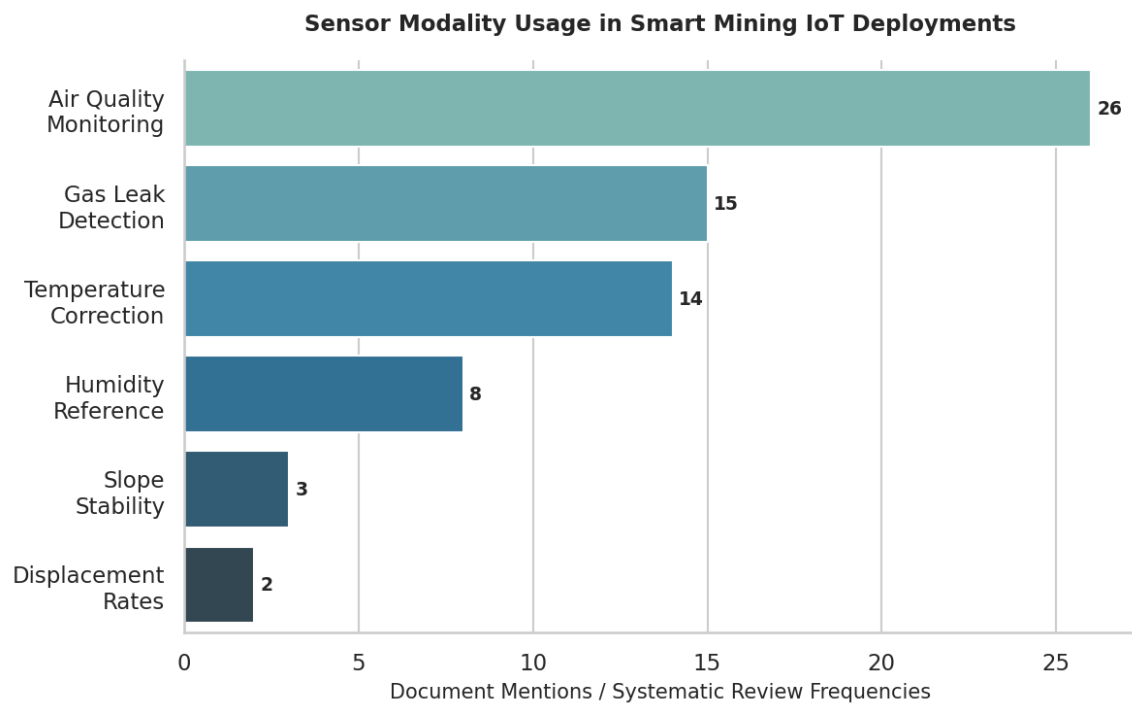
6. Environmental, Gas, and Geotechnics Integration

6.1 DGMS Permissible Gas and Temperature Limits

Surface fractures can serve as direct escape channels for toxic and explosive gases from the underground goaf. The system monitors semiconductor gas sensors against permissible limits defined by the Directorate General of Mines Safety (DGMS).



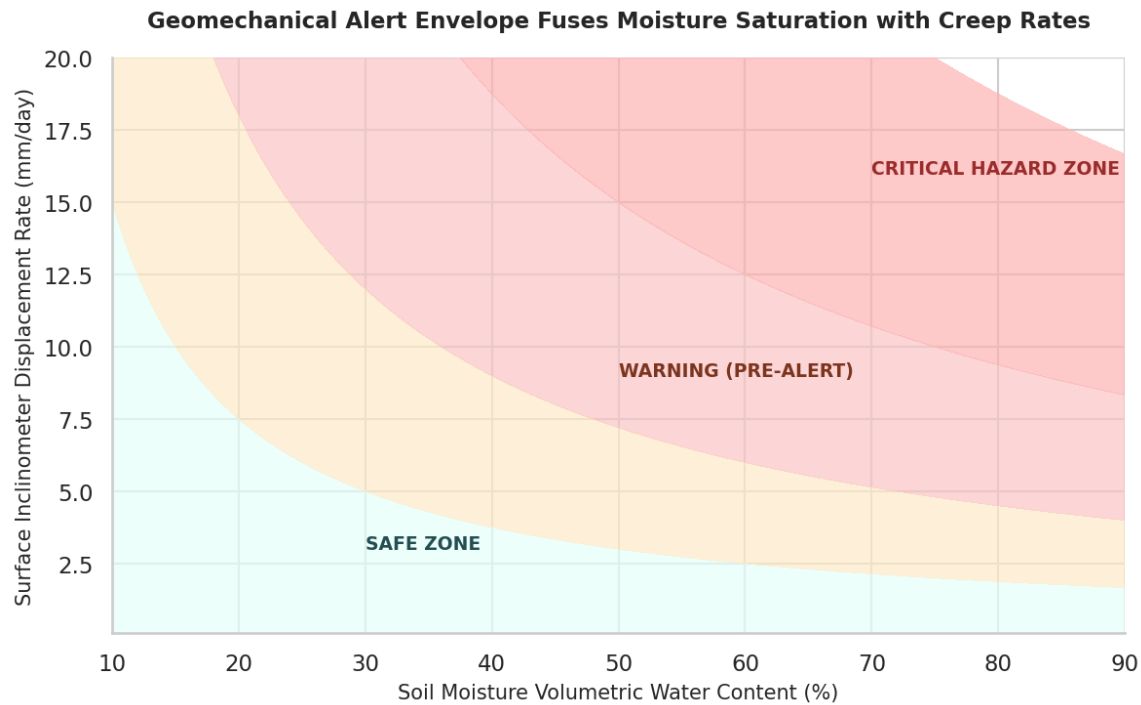
DGMS Permissible Limits and Gas Sensor Thresholds



Quantitative Distribution of Sensor Types in Smart Mining Research

6.2 Bivariate Landslide and Slope Risk Envelopes

To protect steep surface benches and overburden dumps, a bivariate warning matrix plots soil moisture volumetric content against the inclinometer's displacement rate, categorizing the slope state into Safe (Green), Warning (Orange), and Critical (Flashing Red) zones.

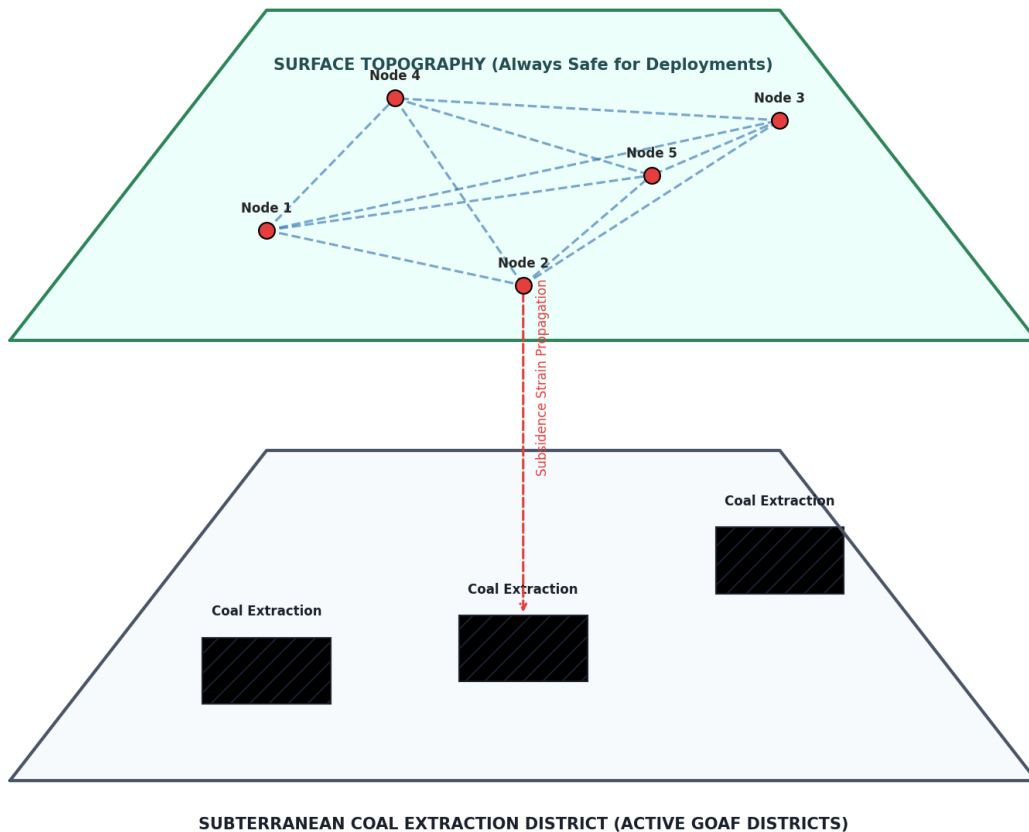


Bivariate Soil Moisture vs. Landslide Risk Envelope

7. Real-Time Command-and-Control Web GIS Dashboard

The web-based dashboard uses Three.js or cesium.js to render a live, interactive 3D terrain mesh. It projects live sensor coordinate offsets (X, Y, Z) onto Digital Surface Models, visualizing vertical subsidence as a dynamic, color-coded heatmap. It serves as a two-way command-and-control interface, allowing operators to monitor node voltage/RSSI and dispatch remote instructions directly to ESP32 nodes.

Surface Sensor Telemetry Array Overlaying Subterranean Active Extraction District



Web GIS Dashboard: AI Engine & Real-Time Alerts Screen

8. Technical and Ethical Audit Framework

Deploying low-cost IoT networks over active mining leases introduces significant ethical responsibilities.

- **The Safety Illusion and False-Negative Accountability:** Deploying uncertified systems creates a false sense of security. The system must be explicitly labeled as a supplementary early-warning tool, not a replacement for statutory manual surveying mandates.
- **Workplace Surveillance vs. Search and Rescue:** While GPS and location tracking are critical life-saving tools during a collapse, they can be co-opted for aggressive labor exploitation and surveillance. Positioning tracking must utilize on-device geofencing that only streams data during structural anomalies or emergency alerts.
- **The Demographic Divide and Academic Exploitation:** Digitization can disenfranchise senior,

highly experienced miners who rely on physical geomechanical cues (like roof sounds). Dashboards must be designed highly intuitively in regional languages to empower older miners. Furthermore, experimental testing often occurs in small, marginalized mines; rigorous ethical reviews must ensure testing equity and direct safety benefits for the hosting test sites.

Conclusion

By shifting telemetry capture entirely to the open ground surface, this system drastically reduces physical complexity, avoids the explosive environments of underground galleries, and secures continuous line-of-sight monitoring over active subsidence zones. Built upon the geomechanical constraints of Naresh C. Saxena's empirical bounds, powered by a self-healing LoRa mesh, and governed by a rigorous ethical framework, this architecture provides a highly scalable and legally compliant roadmap to modernize mine safety in India.